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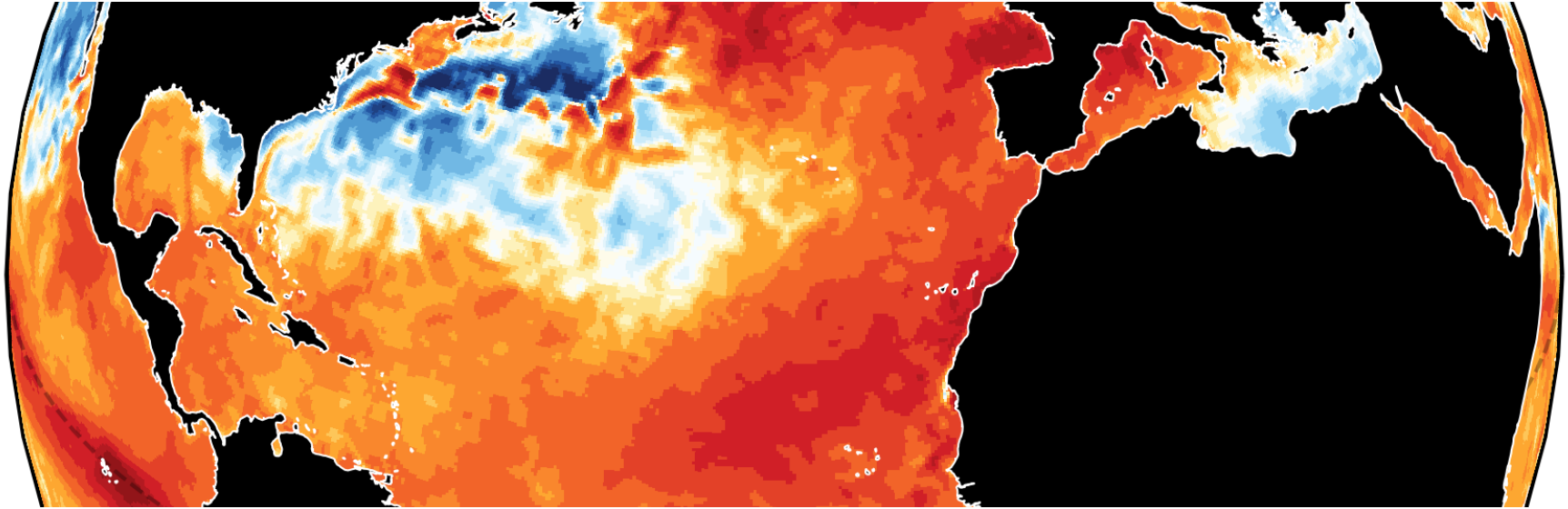

Lea Sophie Dietz

Identifying drivers of Meridional Heat Transport Variability in the Atlantic Meridional Overturning Circulation

Motivation - AMOC variability

- AMOC variability has been misinterpreted as weakening
- No clear trend from observational data alone
- Strong natural variability on seasonal and inter annual timescales

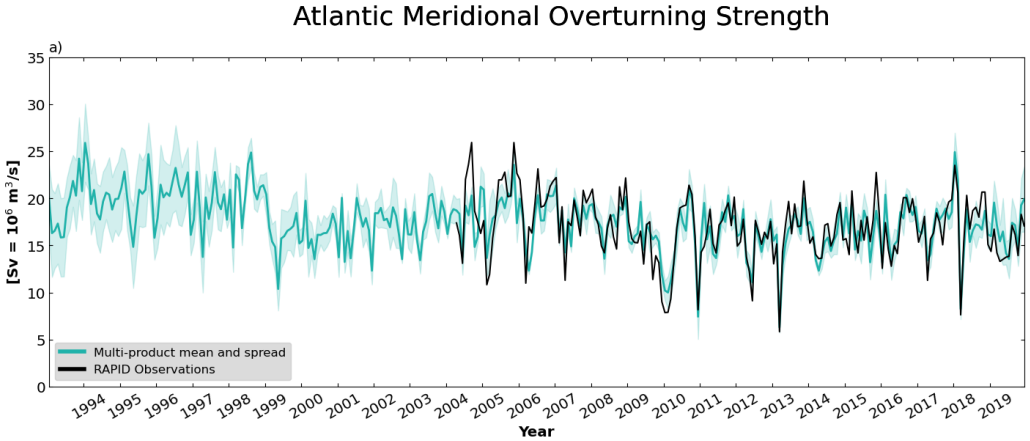

Figure from [2]

- AMOC transport is directly linked with Meridional Heat Transport (MHT)

Motivation - the data challenge

Observational coverage is extremely sparse

Argo Floats
- 1 Float per ~ 85.000 km²

Mooring Arrays
- Profile measurements at specific latitudes
- Cannot capture basin-wide properties

Previous studies mainly used hydrographic data only

Solution: Calafat et. al. dataset - basin wide data

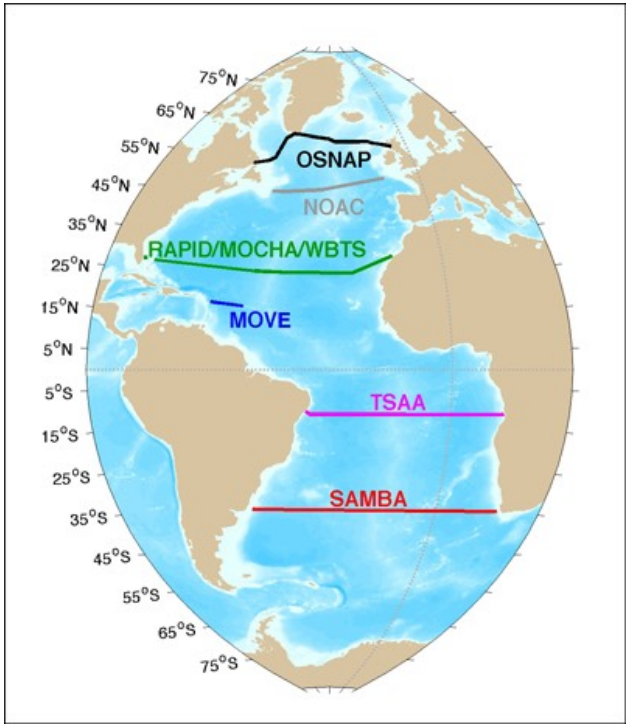

Figure from [3]

Method - data

Dataset from Calafat et. al. [2] combines:

- **Hydrographic data** (mainly argo floats)
  - Thermosteric (TS) and halosteric (HS) height anomalies
  - TS → OHC
- **Satellite altimetry** → sea level (SL)
- **Satellite gravimetry** → ocean mass (OM)

⇒ SL = TS + HS + OM

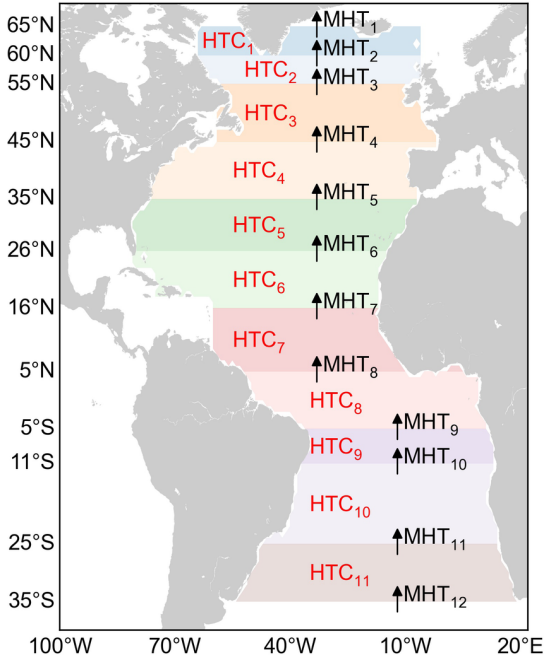

Figure from [4]

Method - data

Estimate **MHT** as the residual of
- surface heat flux (HF) - from Reanalysis
- OHC - from Calafat dataset

Results in basin wide coverage of MHT data:
- **12 latitude bands** from **65°N - 35°S**
- from **2004-2020**

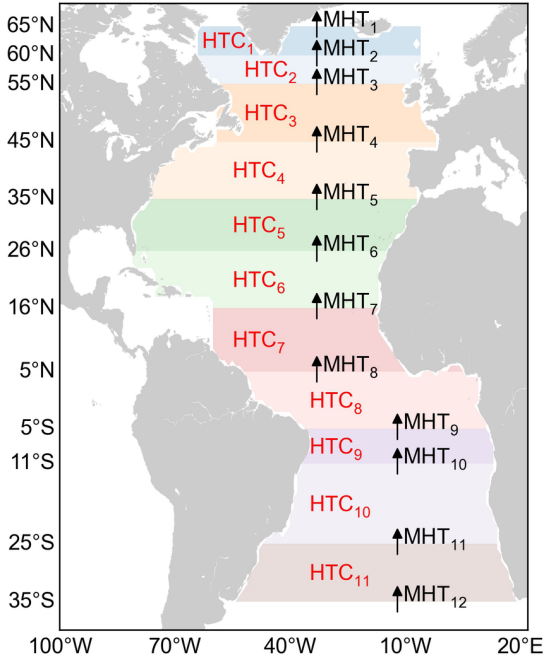

Figure from [4]

Objectives

Using this dataset I want to perform and perform:

Lead/lag correlation analysis: MHT vs. HF at each latitude band

1. **Do surface heat fluxes drive MHT variability or do both respond to common forcing?**
2. **How does MHT variability propagate across latitude bands?**
3. **What are dominant timescales of MHT variability?**

References

1. Johns et. al., „Towards Two Decades of Atlantic Ocean Mass and Heat Transports at 26.5° N“
2. Frajka-Williams et. al., „Compensation between Meridional Flow Components of the Atlantic MOC at 26° N“
3. https://www.o-snap.org/about-us/related-programs/ (all moored AMOC)
4. Calafat et. al., „Estimates of Atlantic Meridional Heat Transport from Spatiotemporal Fusion of Argo, Altimetry and Gravimetry Data“, 2025